CASE STUDY COOPERATIVE AND AUTONOMOUS SYSTEMS · THD · TEAM B3 · KIT 04 · NOVEMBER 2026

How the rover drives from B to I

A LiDAR, ultrasonic and gyro box on the WAVE ROVER that follows a wall, finds a turn landmark, crosses an open square on fixed landmarks, follows a second wall and a railing corridor, and stops at the glass entrance of Block I. No GPS, no camera, rover firmware unchanged.

~120 m
B to I: 20 m wall, 40 m open sea, 10 m arcade, about 50 m interlock and railings

9 logs
mission logs measured with the same code the box runs

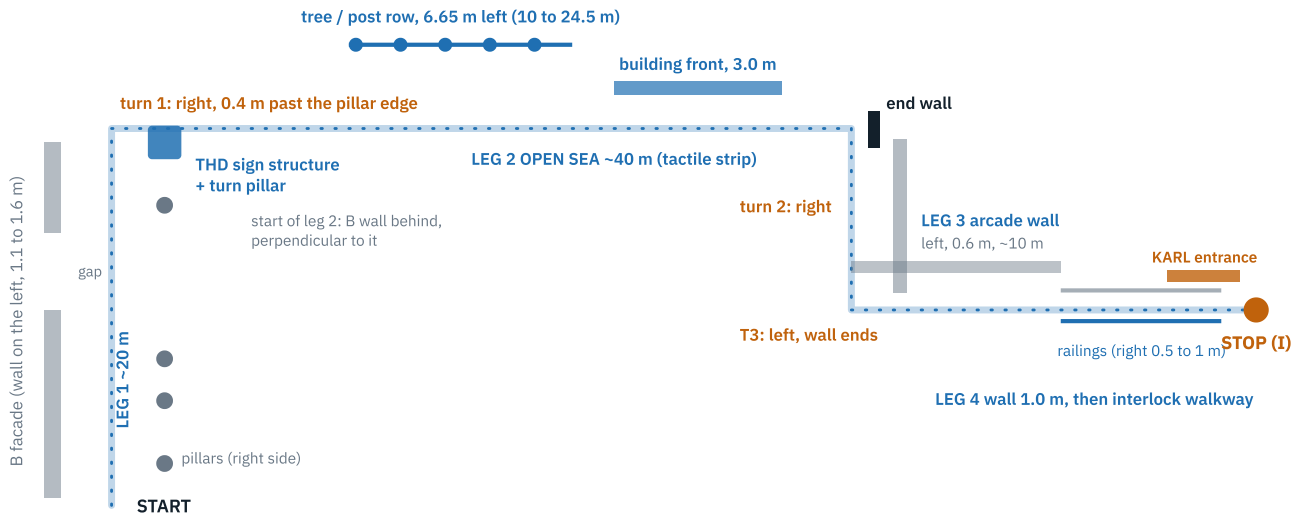
10 versions
each one made after a field run (v1 to v10)

5,670
simulated people cases on the turn, wrong-pillar turns 7.4 % to 1.2 %

In plain words Hardware Software Leg by leg Whose code Logs Simulation Safety Versions Open points References

The route in plain words

Every leg uses a different thing it can see, because no single reference exists along the whole way.



Schematic, not to scale. Blue = what the box steers on in that leg, orange = turns. The camera is not used anywhere.

1. **Leg 1, along the B facade (about 20 m).** The wall is on the left, 1.1 to 1.6 m away. The box keeps that distance and stays parallel to it. Pillars stand on the right. After a gap in the facade the turn pillar comes, with a dark sign structure behind it.
2. **First right turn.** The box finds the sign structure beside the turn pillar, drives 0.4 m past the pillar edge and pivots 90 degrees to the right, perpendicular to the B wall.
3. **Leg 2, the Open Sea (about 40 m).** There are no walls, only a row of trees, stakes and posts on the left, then a building front, then the far end wall. The box steers on those, and on the B wall behind it at the start.
4. **Leg 3 and the left turn T3.** At the end wall it turns right and follows the arcade wall on its left at 0.6 m. When that wall ends (after about 10 m) it turns left around the corner.
5. **Leg 4 to Block I.** It follows the next wall at 1.0 m, crosses the light interlock walkway, enters the corridor between two railings, climbs the slight ramp and stops 1.5 m after the recessed glass entrance (KARL).

In one sentence: the rover reads the shape of the campus with its LiDAR, uses the gyro to hold a straight line in between, and stops and waits whenever something is in front of it.

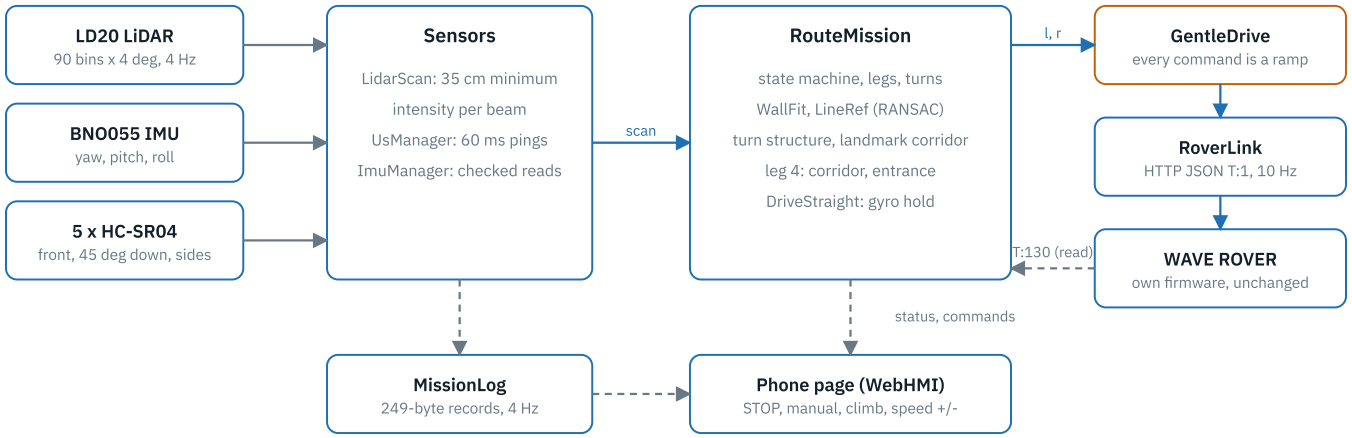
Hardware and what each sensor is used for

Numbers from the datasheets (checked online during this work) next to what we measured in the field.

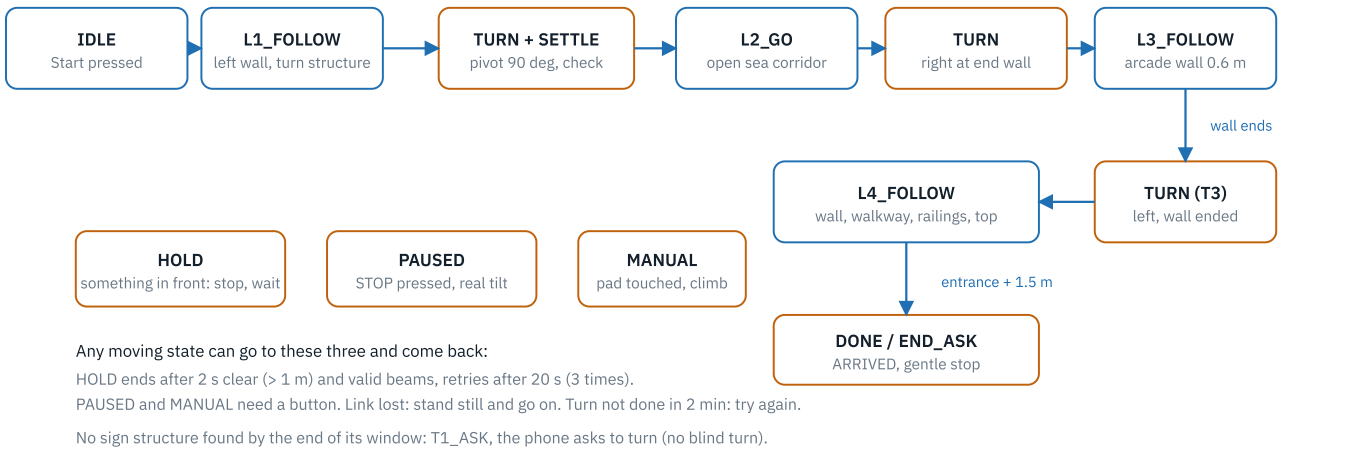
PART	DATASHEET	WHAT WE MEASURED	USED FOR
LDROBOT LD20 LiDAR (LD19 family)	0.02 to 12 m, 4500 samples/s, 5 to 13 Hz scan, 30 klux ambient light, class I. LD20 has no datasheet we could find; the LD19 one is used.	We keep 90 bins of 4 degrees, 4 Hz. Mounted 7 degrees rotated: offset 83 degrees. In daylight 15 to 21 % of the scans had ghost points at 20 to 34 cm (night: 0 to 5 %).	Wall fits, landmark corridor, structure, end wall, obstacles
5 x HC-SR04 ultrasonic	2 to 400 cm, about 15 degree cone, measuring cycle above 60 ms	Pings 60 ms apart. Front sensor floor echo at 50 to 90 cm (40 % of readings with nothing ahead). One front sensor is angled 45 degrees down; the rear one is broken (reads 4 cm).	Verifies near LiDAR obstacles, stops at under 35 cm
Bosch BNO055 IMU	9 axis fusion (NDOF). Heading drift in NDOF comes mostly from magnetic distortion; the fusion library (BSX) is not public.	Gyro heading drifts about +-2 degrees per minute at random in the open sea. Pitch and roll for tilt stop, ramp and edge detection.	Heading hold, turns, pitch rules
ESP32 box (ESP32 Dev Module)	2 cores, Wi-Fi access point	Program 96 % of flash.	Runs everything, serves the phone page, logs 249-byte records at 4 Hz
WAVE ROVER	JSON over HTTP: T:1 speed, T:130 status	10 Hz commands, 0.5 s deadman. No wheel encoders: odometry is the commanded speed times time.	The only actuator. Its firmware is never changed.

Software: what runs where

One state machine, a few small modules, and every command to the wheels passes through the same ramp stage.



Data flow in the box. GentleDrive is the last stage before the rover: every change of the wheel command is a ramp, which protects the gearbox.



State machine of RouteMission.cpp. HOLD (something in front), PAUSED (STOP or tilt) and MANUAL (pad touched) can interrupt every moving state.

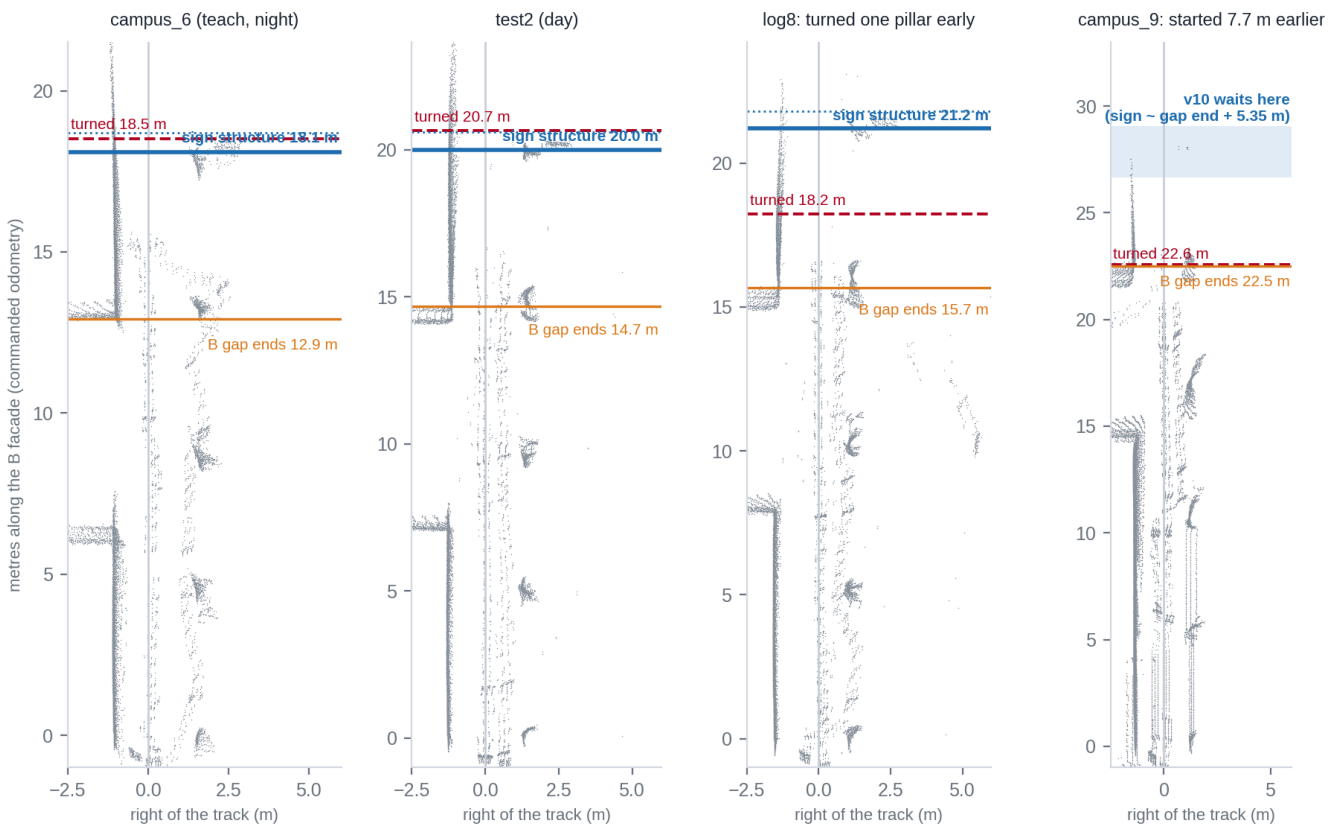
FILE	JOB
`RouteMission.cpp/.h`	State machine, leg logic, turn decision, open-sea references, leg 4, edge assist, status for the page (about 1,500 lines)
`WallFit.h`, `LineRef.h`	Wall line fit in the side frame, RANSAC line fits for sparse landmark rows and far walls, perpendicular heading
`DriveStraight.h`	Gyro heading hold with learned wheel trim and a 12 degree failsafe
`GentleDrive.h`	Ramps for speed, turns and single wheels (turn mode)
`LidarScan.h`, `UsManager.cpp`, `ImuManager.cpp`, `Sensors.cpp`	LD20 packets to 90 bins with intensity and 35 cm minimum range; ultrasonic ping scheduling; IMU reads
`RoverLink.cpp`	10 Hz JSON link to the rover in its own task, deadman, read-only status
`MissionLog.cpp`	Flash log with all sensors, outputs and 67 event codes
`MissionPage.h`, `WebHMI.cpp`	Phone dashboard: STOP, manual pad, climb, speedometer, speed override, settings, log download
`tools/` (Python)	`decode_mission_log.py`, `narrate_run.py` (what the rover thought), `landmark_benchmark.py` and `landmark_db.json`, `fwmirror.py` (Python copy of the firmware logic), `replay_turn.py`, `sweep_start.py`, `sim_open_sea.py`, `sim_people.py`

Leg by leg: reference, reason, code and logs

For each part of the route: what the box looks at, why that and not something else, where the code is, and which logs supported it.

Leg 1: left wall of B, and the turn

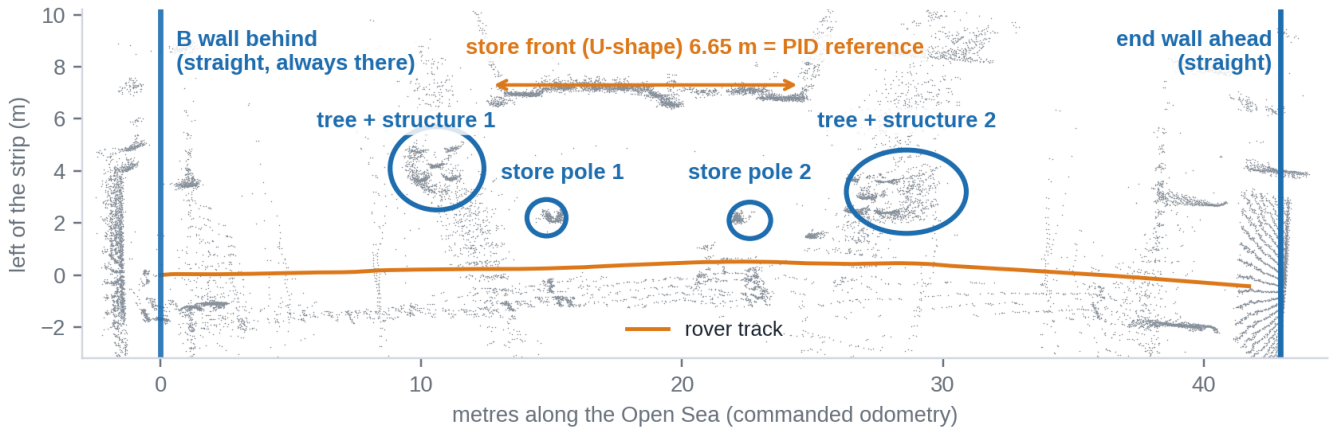
REFERENCE	WHY	CODE	EVIDENCE
Left facade line (WallFit), keep the distance it had at the start	Long straight wall at 1.1 to 1.6 m; held 1.57 to 1.59 m for 20 m once the LiDAR 7 degree rotation was removed	`_leg1`, `WallFit::measure`	campus_2, campus_3, campus_5, campus_6, log8, campus_9
Turn structure: the dark sign structure behind the turn pillar, found in an evidence grid fed only with beams ahead and abeam	log8 turned one pillar early because a person behind the rover merged with an ordinary pillar into a wide landmark. Ordinary pillars have nothing behind them. The structure is fixed.	`_structAdd`, `_structDetect`, `_structUpdate`, `_structChoice`	Found in all 7 runs that reach the turn
B-gap anchor: the left wall comes back after the gap, the sign is 5.35 +- 1.2 m later	campus_9 started 7.7 m earlier and the fixed 15 to 25 m window missed the sign. The distance from the gap end is the same in every run.	`_structWin`, gap anchor in `_leg1`	log8 15.7 to 21.2 m, test2 14.7 to 20.0, campus_6 12.9 to 18.1, campus_5 13.7 to 19.2, test1 15.2 to 20.5
Turn 0.40 m past the pillar edge, target B wall direction minus 90 degrees (as in v5)	This timing put the crossing on the strip in the runs that worked	case `L1_FOLLOW`, `_landmark` (Kethan's turnLandmark, widened)	Replay: campus_6 18.7 m (v8 18.51), test2 20.6 (20.66), campus_5 19.8 (19.82)



Leg 1 seen from above in four runs (LiDAR points, commanded odometry). In log8 the red line (turn) is one pillar before the blue line (structure); in campus_9 the rover turned at the gap end, 5.7 m before the sign. v10 waits in the shaded window.

Leg 2: the Open Sea, landmark corridor

REFERENCE	WHY	CODE	EVIDENCE
B wall behind (0 to 14 m): perpendicular heading	Flat, long, close at the start	`case L2_GO`, `LineRef::sectorFit`, `perpHeading`	Perpendicular within +-1.3 degrees in 6 runs
Tree / post row on the left, 6.65 m, used from 10 to 24.5 m	No wall to match here; the row is fixed and straight enough. The gyro alone drifts about +-2 degrees per minute at random.	`LineRef::rowFit`, PID kp 10, ki 1.5, kd 20, limit 8 degrees	Offset 6.5 to 6.8 m in 5 of 9 runs (teach run campus_6 6.67), seen on 39 to 56 % of the scans
Building front on the left, 3.0 m, 26 to 40 m	The row ends; the building takes over the side position	same	seen in log8, test2
End wall ahead, after 85 % of the crossing; **flat and wide**, leaky counter	A person or a small group in front is flat but not wide; real end walls stay within 0.3 m over +-48 degrees	`_frontWall` (Kethan's frontWallWithin), `_frontWallWide`	Real end walls at 37.8 to 49.1 m in 6 logs, all accepted



Open Sea of campus_9 in the strip frame. The rover track is the orange line; the blue lines are the four landmarks the box steers on.

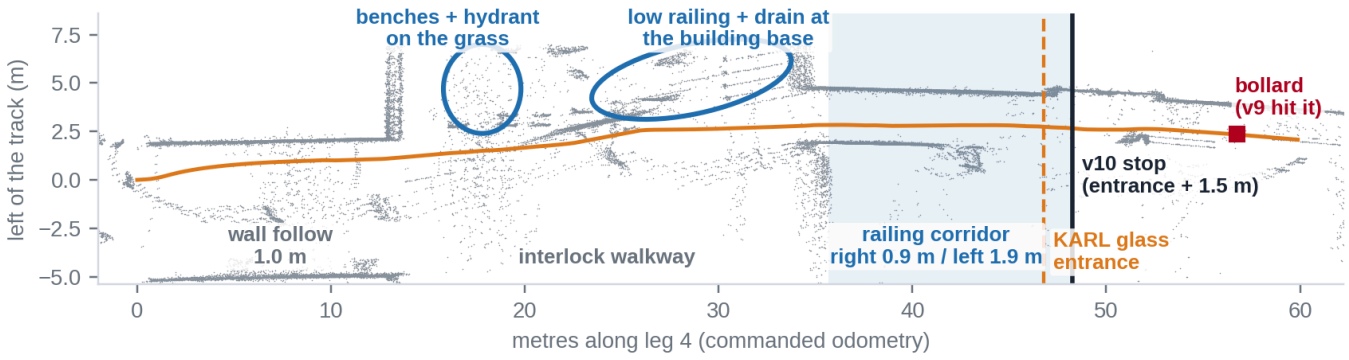
Known weakness. The row steers the side position only until 24.5 m. With a gyro bias of 6 degrees per minute the simulation ends 0.4 m off at the end wall (the earlier version had 0.06 m). The simulated worst mix of drift, slide and missing trees ends 0.41 m off at 24 m (before the PID: 1.02 m).

Leg 3 and T3: the arcade wall

Same wall follower as leg 1, closing in to 0.6 m at 20 degrees per metre (never more than 10 degrees ahead of the heading, 2 degrees per scan: no failsafe in the simulation). When no wall is seen for 1.5 m it turns left, 90 degrees from the wall direction. Logs: campus_5, campus_6, test2, log8, campus_9 (arcade length 8.3 to 10.5 m).

Leg 4: to Block I

PHASE	REFERENCE	CODE	EVIDENCE
Wall follow, 1.0 m, power 0.30	Forward-left line within 15 degrees of the arcade direction plus 90	`_leg4Wall`	campus_6, test2, log8, campus_9
Interlock walkway	Wall direction, left edge at its first distance (grass, strip, bench)	`_l4Walk` phase 1	log8 manual drive, campus_9
Railing corridor	Right line 0.35 to 1.3 m and left facade 1.4 to 2.6 m, together 2.4 to 3.3 m, 1 s	`_l4Walk` phase 2	log8: 40.0 m, campus_9: 35.6 m, never earlier
KARL entrance, stop	Left facade steps back (1.7 to 2.0 m) and glass drops beams; stop 1.5 m later, gentle ramp	`_l4Walk` , case `L4_FOLLOW`	campus_9: 47.0 m, stop 48.5 m, bollard 56 m
Edge assist	Pitch stays 3 to 12 degrees above its recent minimum for 3 s with nothing within 1.2 m	`_edgeAssist`	log8: 60 s stuck at the cobble to interlock edge at 0.22



Leg 4 of campus_9. v9 passed the entrance and hit the bollard; v10 stops 1.5 m after the entrance (replay of the detectors on this log).

Whose code is what

Team code the box builds on is credited in the source comments. Everything in the second table was developed in this work.

FROM THE TEAM, USED AS A BASE (CHECKED FILE BY FILE AGAINST THE TEAM'S CODE, 10 NOV)	WHERE IN THE BOX CODE
Kethan: gyro heading hold (PD) and its ~0.06 s yaw-rate filter, the "step to full power" turn, the pivot release before the target (stop at 83 of 90 deg), `turnLandmark`, `frontWallWithin`, the field-tested IMU read version, the drive log (`DriveLog`)	`DriveStraight.h`, `GentleDrive.h`, `ImuManager.*`, `RouteMission.cpp` (`_turn`, `_landmark`, `_frontWall`), `MissionLog.*` (built on `DriveLog`)
Team box base (Prakash's `cas_box` repository, also in Kethan's copy): sensor manager, ultrasonic, GPS, display, web page server, rover link, LiDAR scan, the wall-line fit `WallFollow::measure`	`Sensors.*`, `UsManager.*` (1–7 lines changed), `GpsManager.*`, `DisplayManager.*` (identical), `WebHMI.*`, `RoverLink.*`, `LidarScan.h` (35 cm minimum range, ghost filter added), `WallFit.h` (unchanged)
Prakash: `SafetyGuard` (collision avoidance layer 1: tilt stop, front ultrasonic / LiDAR stop with turning still allowed, rear ultrasonic), `HeadingHold`, `RouteFollow` / `RouteLocalizer` (route follower, local path)	not linked into this build. The same safety ideas are implemented in the mission itself: emergency stop at 0.45 m, ultrasonic stop at 35 cm, hold and wait for people, tilt pause
Manikant: collision avoidance and an Open Sea approach with SLAM, shown in the team slides	no source code of it is in the shared folders (the `mani/` folder holds Kethan's logging code and logs), so nothing of it is used here

NEW IN THIS WORK	VERSION	WHERE
Wall follower with new-wall acquisition, bend acceptance, separate scan and good timers	v4	`_leg1`
Turn tactics cycle (pivot, swing, shift, arc) with per-wheel ramps, stall detection	v3	`_turn` , `GentleDrive`
Landmark corridor and cross-track PID in the Open Sea	v6, v7	`case L2_GO` , `LineRef.h`
Leg 3 and leg 4 wall approach law	v7, v8	`_leg4Wall` , `_leg1`
Daylight ghost filter: 35 cm minimum range, near obstacle verified by 4 beams or the ultrasonic	v8, v8.1	`_front` , `LidarScan.h`
No phone heartbeat, automatic recovery (link wait, retry, realign); only STOP and tilt pause	v8	`update` , case `HOLD`
Turn structure with B-gap anchor	v9, v10	`_struct*`
Speedometer, speed override, climb button, record button	v8, v9	`MissionPage.h`
Wide end wall, leaky counter, hold release for vanished obstacles	v10	`_frontWallWide` , case `HOLD`
Railing corridor, KARL entrance, edge assist	v9, v10	`_l4Walk` , `_edgeAssist`
Ramp opening finder in leg 4 (cross line + facade / railing corner, wait instead of entering), sideways positioning after a late turn before the Open Sea, backwards shift in a blocked turn, no turn before the B-gap end + 4.5 m	v10.3	`_l4FindGap` , `_l4Gap` , `_afterTurn` , `_turn`
Log decoder, narrator, landmark benchmark, Python mirror of the firmware, replay, people and weather simulation	all	`tools/`

Logs used and why

Every field run produced one mission log (flash on the box) and usually a phone video. The log was decoded record by record before any change.

RUN	DATE	WHAT HAPPENED	USED FOR
`campus_2`	7 Nov	First full leg 1 and turn; the full pivot locked at the turn spot after 4 to 7 degrees	Turn tactics, turn landmark, end wall level check
`campus_3`	7 Nov	Entrance board test; wall-lost never fired (timer bug)	Wall follower v4, converging wall
`campus_4` , `campus_5`	7 Nov	Gyro drift correction pulled the rover off the strip	Removed drift correction, first reference types
`campus_6` (night) `teach`	7 Nov	Best run; tree row 6.67 m, strip direction	Landmark offsets, PID design, simulation world
`day_test1` , `day_test2`	8 Nov	Daylight: ghost points at 20 to 34 cm; false end wall in the open sea (test1)	Ghost filter, hold end-wall bug
`log8`	8 Nov	Turned one pillar early (follower behind); first interlock drive with the climb button	Turn structure, leg 4 corridor, climb and edge behaviour
`campus_9`	8 Nov	Turned at the gap (start 7.7 m earlier); open sea and leg 4 fully autonomous; hit a bollard after the entrance	B-gap anchor, corridor width, entrance, hold release

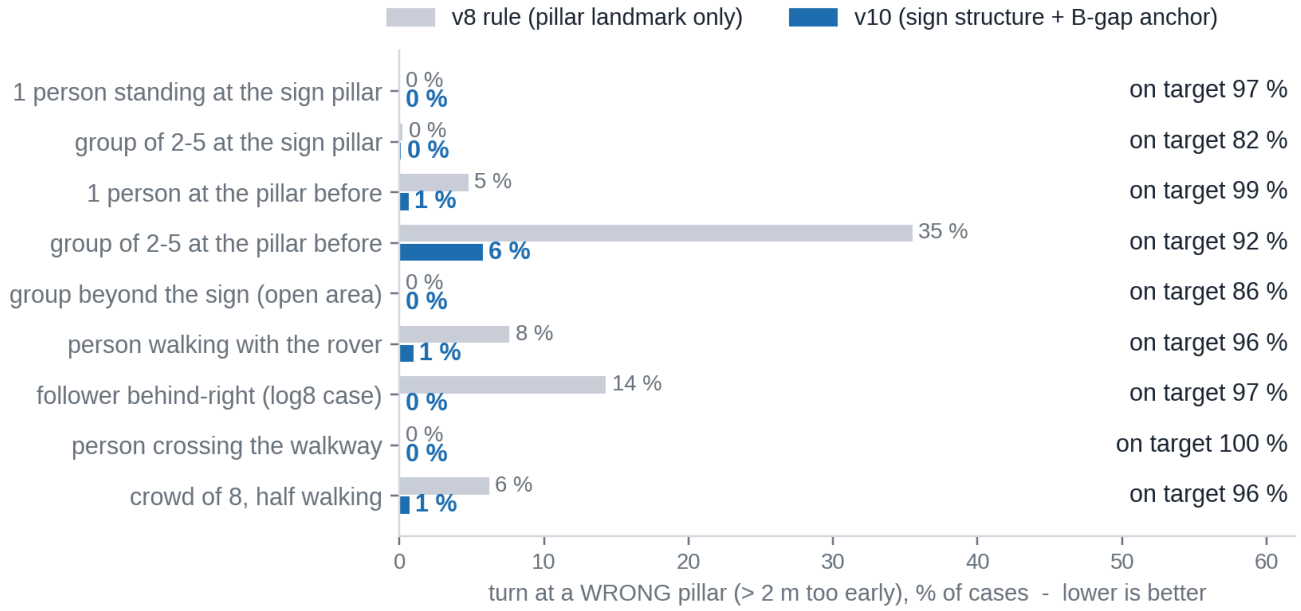
Why these and not others: the night teach run is the cleanest reference, the daylight runs show what changes with sun, log8 and campus_9 are the only runs that reach the end of the route. Runs with a person touching the rover or a heartbeat pause (campus_2 to campus_5) were used to find bugs but never to tune numbers.

Simulation: people, crowds and weather

Every change was replayed on the recorded runs and simulated before it went into the box.

People near the turn pillar

People (circles of 0.15 to 0.30 m radius) are ray-cast into the real leg 1 scans of seven runs, standing, walking along, towards, across, a follower behind, groups of 2 to 5 and a crowd of 8. The same turn logic as in the box then decides. A turn more than 2 m early counts as a wrong pillar.



Wrong-pillar share per scenario (lower is better) and the share of turns within +-0.5 m of the clean turn. Totals: v10 66 of 5,670 wrong (1.2 %), 93.5 % on target; v8 rule 421 wrong (7.4 %).

Open Sea: crowds (20 random starts each) and weather

SCENARIO	FALSE END WALL, OLD CHECK	FALSE END WALL, V10
1 person standing in the path (last 30 %)	18 of 20	0 of 20
Group of 3 standing in the path	20 of 20	0 of 20
Walking crowd of 20	13 of 20	0 of 20
Row of 8 across at 30 m	20 of 20	0 of 20
12 standing people, crowd at the tree row, group across at 15 m, people walking along	0 to 2 of 20	0 of 20

ENVIRONMENT (NO PEOPLE)	REACH THE END WALL	FALSE END WALL	FALSE HOLDS
Clear night	12 of 12	0	0
Daylight ghosts, 20 % of scans	12 of 12	0	0
Low sun, 40 degree wedge blind in 30 % of scans	12 of 12	0	0
Light rain (3 % random near returns, 10 % dropouts)	12 of 12	0	0
Heavy rain or snow (8 % near returns, 25 % dropouts)	10 of 12	0	0
Wet slip 8 % and gyro -6 degrees per minute	12 of 12	0	0
Worst case: snow, slip, drift and sun together	7 of 12	0	1

Open points. Groups of 3 to 5 around the pillar before the sign still cause 6 % wrong turns. Eight people shoulder to shoulder at the end wall are not covered. In the worst combined case the rover drifts about 2.5 m off the strip because the row is lost. All of this is replay and simulation: v10 has not been driven yet.

Start position sweep

Each log was shifted by -8 to +8 m along B. The turn moved by 0.00 to 0.13 m in all seven runs, because the window is measured from the gap end.

Safety rules that never change

- **Turns need full power** (0.50, a step to full like Kethan's code); forward driving is slow (0.22, leg 4 0.30).
- **Every stop is a ramp.** Normal 0.30 per second, obstacle 0.80 per second, never a jump to zero.
- **People in front: stop and wait.** No swerving. After a hold the rover lines up again and retries.
- **Only STOP and a real tilt pause.** A lost rover link makes it stand still and continue.

- **The rover firmware is not touched.** The box only sends T:1 and reads T:130 and T:139.

- **The ESP32 is not stressed.** No flash erase from the program, no heavy Wi-Fi traffic.
- **Flash at 115200**, never Erase All Flash.
- **Obstacles are verified.** A near LiDAR point needs 4 beams or the front ultrasonic; an obstacle that vanishes from closer than 0.45 m is still treated as there.

From v1 to v10

VERSION	TRIGGERED BY	CHANGE
v3	campus_2	Wall follower on leg 1, turn tactics cycle, leg 3, level check for the end wall; camera removed from the route
v4	campus_3	Wall follower timers and new-wall acquisition, entrance board
v5	campus_4, campus_5	Reference types in the open sea, turn landmark edge tracking (pillar edge + 0.40 m)
v6	campus_5 + video	Landmark corridor: tree row and building front
v7	campus_6 teach run	Cross-track PID, leg 4 wall gate
v8, v8.1	day_test1, day_test2	Daylight ghost filter, ultrasonic verification, no heartbeat, climb and record buttons, leg 3 at 0.6 m
v9	log8	Turn structure, speedometer and speed override, leg 4 corridor and ramp, edge assist, wide end wall
v10	campus_9	Start-independent turn window, no blind turn, hold release, KARL entrance stop, end wall gate 85 %

Open points and next field test

- Drive v10 on the whole route and check: turn at the sign pillar from a different start, stop at the KARL entrance.
- Open sea: the side position is steered only until 24.5 m; a gyro bias then moves the rover about 0.4 m by the end wall.
- Crowds: groups of 3 to 5 at the pillar before the sign; shoulder-to-shoulder rows at the end wall.
- No wheel encoders: all distances are commanded odometry. Slip on wet stones makes them too long.
- Test the crowd cases for real, for example with team members standing at the sign pillar.

References

- LDROBOT LD19 / STL-19P datasheet (ranging 0.02 to 12 m, 4500 Hz): Waveshare file; kit pages Waveshare DTOF LIDAR LD19, DFRobot LD19 manual
- HC-SR04 datasheet (2 to 400 cm, 15 degrees, 60 ms cycle): HCSR04 manual, classic datasheet
- Bosch BNO055 NDOF heading behaviour: Bosch Sensortec community thread
- Team slides: Field Tests log_009 to log_011, local path tracking log_020 to log_022 (steering, collision avoidance and SLAM by other team members).
- Project files: `deliverables/v10_BOX`, `cas_box_mission/tools`, `PPT/new_slides_assets`.

Prepared from the mission logs in `Log_run/`. Simulation numbers are from `tools/sim_people.py` and `tools/sim_open_sea.py`. The LD20 specification was not found; LD19 figures are used for it.